

HANO, SwedenWMO: **026280**Lat: **56.0137N**Lon: **14.8462E**Elev: **60**StdP: **100.61**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.1	-5.6	-11.1	1.5	-5.1	-9.4	1.7	-4.1	19.7	2.5	18.3	1.9	8.0	40	0.812

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	23.7	18.7	22.3	18.2	21.0	17.5	19.9	22.2	19.0	21.3	18.1	20.4	5.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	21.4	18.0	13.1	20.4	17.1	12.3	19.6	57.3	22.4	54.2	21.5	51.5	20.4	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	15.4	14.0	-8.9	26.4	2.2	1.9	-10.4	27.7	-11.7	28.8	-12.9	29.9	-14.5	31.2
			-9.5	20.5	2.2	1.2	-11.1	21.4	-12.3	22.2	-13.6	22.9	-15.1	23.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	0.8	0.7	2.4	5.8	10.5	14.4	17.3	17.2	13.8	9.2	5.4	2.4	
	DBStd	6.59	3.19	3.04	2.82	2.43	2.62	2.38	2.53	2.37	2.30	2.66	2.83	3.39	
	HDD10.0	1352	286	259	234	128	24	1	0	0	1	46	138	236	
	HDD18.3	3669	544	493	493	377	242	119	51	51	137	282	387	494	
	CDD10.0	763	0	0	0	2	41	134	227	223	114	22	1	0	
	CDD18.3	38	0	0	0	0	0	2	20	16	1	0	0	0	
	CDH23.3	61	0	0	0	0	0	4	39	17	0	0	0	0	
	CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0	
Wind	WSAvg	7.3	8.4	8.2	7.6	6.8	6.5	6.1	5.8	6.3	7.1	7.9	8.2	8.5	
Precipitation	PrecAvg	594	45	35	28	32	43	56	64	58	57	60	64	51	
	PrecMax	790	81	76	65	64	144	156	220	146	167	126	122	106	
	PrecMin	453	0	9	4	3	7	0	4	4	9	10	16	21	
	PrecStd	82	22	18	14	17	26	32	40	33	38	30	25	19	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	8.2	11.4	14.4	19.9	23.7	26.4	25.2	21.2	15.4	12.2	9.2	
		MCWB	6.8	6.4	7.5	9.7	13.9	17.2	19.0	19.7	18.0	13.7	11.0	8.1	
	2%	DB	6.4	6.4	8.6	12.3	17.5	21.1	24.4	23.2	19.5	14.2	10.9	8.3	
		MCWB	5.5	5.2	5.6	8.5	13.3	15.9	19.0	18.9	16.3	12.8	9.4	7.3	
	5%	DB	5.5	5.4	7.1	10.8	16.0	19.6	22.8	22.0	18.1	13.4	9.8	7.4	
		MCWB	4.7	4.3	4.9	8.0	12.3	15.2	18.4	18.4	15.4	12.1	8.5	6.6	
	10%	DB	4.7	4.5	5.9	9.5	14.6	18.3	21.4	20.8	17.0	12.6	9.0	6.6	
		MCWB	4.0	3.6	4.4	7.0	11.4	14.7	17.9	17.7	14.7	11.5	8.0	5.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	7.0	8.3	10.8	15.5	18.5	21.2	20.9	19.0	14.4	11.5	8.6
			MCDB	7.3	7.6	10.8	13.5	18.0	21.7	23.8	23.6	21.0	14.9	11.9	9.0
		2%	WB	5.8	5.5	6.2	9.3	14.1	17.0	20.2	19.7	17.0	13.2	10.0	7.5
			MCDB	6.3	6.1	7.6	11.5	16.4	19.5	22.7	22.0	18.6	13.9	10.6	8.0
5%		WB	4.9	4.5	5.3	8.2	13.0	16.1	19.2	19.0	16.0	12.4	8.9	6.8	
		MCDB	5.3	5.2	6.5	10.2	15.5	18.5	21.7	21.1	17.5	13.1	9.6	7.3	
10%		WB	4.1	3.7	4.5	7.2	11.9	15.2	18.3	18.2	15.2	11.7	8.2	6.0	
		MCDB	4.6	4.4	5.6	9.1	14.1	17.6	20.8	20.2	16.5	12.5	8.8	6.5	
Mean Daily Temperature Range		5% DB	MDBR	2.7	3.0	4.1	4.8	5.3	5.6	5.6	4.9	4.3	3.2	2.6	2.7
			MCDBR	2.7	3.7	6.0	6.9	7.1	7.5	7.2	6.0	5.4	3.6	2.7	2.9
	MCWBR		2.7	3.0	4.0	4.2	4.1	4.1	3.7	3.3	3.1	2.8	2.8	2.8	
	5% WB	MCDBR	2.6	3.4	5.4	6.4	6.2	6.3	6.1	5.1	4.5	3.2	2.7	2.8	
		MCWBR	2.7	3.0	4.0	4.2	4.2	4.0	3.5	3.3	3.1	2.7	2.8	2.9	
Clear-Sky Solar Irradiance	taub	0.309	0.334	0.353	0.386	0.384	0.376	0.399	0.400	0.369	0.356	0.336	0.301		
	taud	2.292	2.322	2.334	2.313	2.375	2.427	2.381	2.381	2.446	2.403	2.332	2.266		
	Ebn at Noon	593	726	808	832	857	868	837	810	787	693	553	501		
	Edh at Noon	53	75	94	111	111	107	110	103	85	70	52	43		
All-Sky Solar Radiation	RadAvg	0.41	1.02	2.46	4.17	5.49	5.85	5.34	4.31	3.01	1.42	0.54	0.29		
	RadStd	0.08	0.18	0.24	0.47	0.54	0.37	0.62	0.36	0.33	0.24	0.10	0.03		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-162	N/A	N/A

Nomenclature: See separate page